THE NOVA RESEARCH CURRICULUM

Chemistry

Grade 7 · Term 1

Student edition

Outcome: CH-7.1 atomic structure, electron shells, valency and chemical formulae.

Lesson notes

- **Atomic structure (Weeks 1–4):** protons (+) and neutrons (no charge) in the nucleus; electrons (−) in shells. Atomic number = number of protons. Atoms are neutral: protons = electrons.

  ▶ **Try it on NovaSchool:** Building atoms: electron shells — open **Simulations** on your dashboard.

- **Shells and valency (Weeks 5–8):** shells fill 2, 8, 8 for the first 20 elements. Valency = electrons an atom gains, loses or shares to fill its outer shell: H 1, Na 1, Cl 1, O 2, Mg 2, Al 3, C 4.
- **Formulae (Weeks 9–12):** H_2O, CO_2, NaCl, MgO; criss-cross method with valencies: Al (3) and O (2) → Al_2O_3. Name simple compounds (sodium chloride, magnesium oxide).

Workbook pages

Page	Task
1–2	Draw shell diagrams for elements 1–20.
3	Valency table.
4	Write 12 formulae with the criss-cross method.

Question bank

#	Outcome	Question	Your answer
1	CH-7.1	Which particle has a negative charge?	proton / neutron / electron / nucleus
2	CH-7.1	Atomic number equals the number of …	neutrons / protons / shells / molecules
3	CH-7.1	Sodium has 11 electrons. Its shell arrangement is …	2, 9 / 2, 8, 1 / 8, 2, 1 / 2, 7, 2
4	CH-7.1	The first shell can hold … electrons.	2 / 8 / 18 / 1
5	CH-7.1	Valency of magnesium?	1 / 2 / 3 / 4
6	CH-7.1	Formula of aluminium oxide?	AlO / Al_3O_2 / Al_2O_3 / AlO_3
7	CH-7.1	An atom has equal numbers of protons and electrons. True or false.	True / False
8	CH-7.1	What does H_2O tell us?	_____